

# Instacart Market Basket Analysis

*Understanding customer ordering behavior to guide business decisions*

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## Problem Statement

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A grocery ordering platform generates a large volume of order data every day, but that data on its own doesn't tell the business anything useful. before any decision gets made on inventory, staffing, or retention spend, a few basic questions need real answers: which products and categories actually drive order volume and deserve priority, how often customers order and how big their baskets tend to be, when demand genuinely peaks during the week, how much of the business runs on repeat purchases versus first-time purchases, and which categories deserve the most shelf space and marketing attention.

This report works through those questions using close to 32 million order-item records from the Instacart dataset. It moves directly from what the data shows to what that means for the business, and closes with a short set of recommendations that follow from the findings, not from assumption.

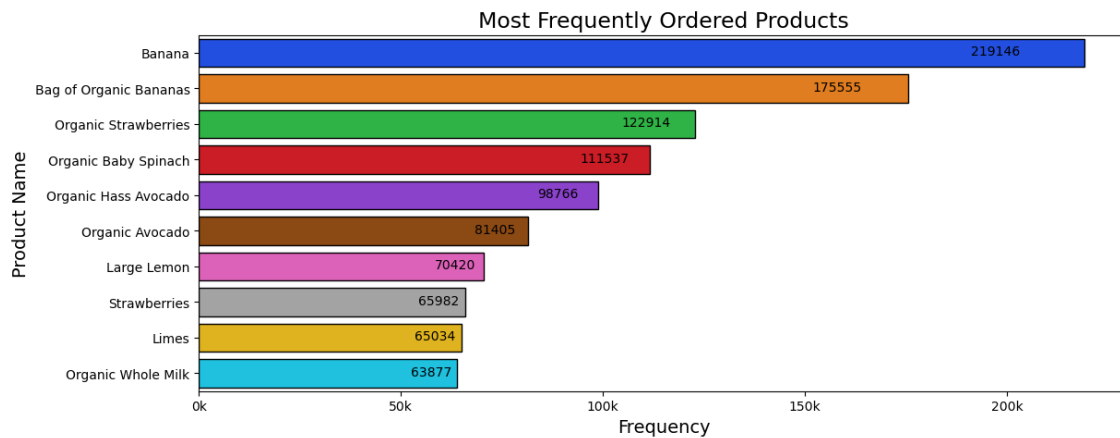
| Metric                            | Value                                           |
|-----------------------------------|-------------------------------------------------|
| Reorder share of all items        | 59% (vs. 41% first-time)                        |
| Top product / department          | Banana (219,146 orders) / Produce (4.39M items) |
| Highest product reorder rate      | Whole Organic Omega 3 Milk — 86%                |
| Peak order day / window           | Sunday (2.87M orders) / 9 AM – 4 PM             |
| Average basket size / order cycle | 10 items / ~13 days between orders              |

## Insights

The findings below are grouped by theme: what people buy, how often they buy it, when they buy, and how loyal that buying behavior actually is. Each one is tied back to what it means for the business, not just what the chart shows.

## What People Buy

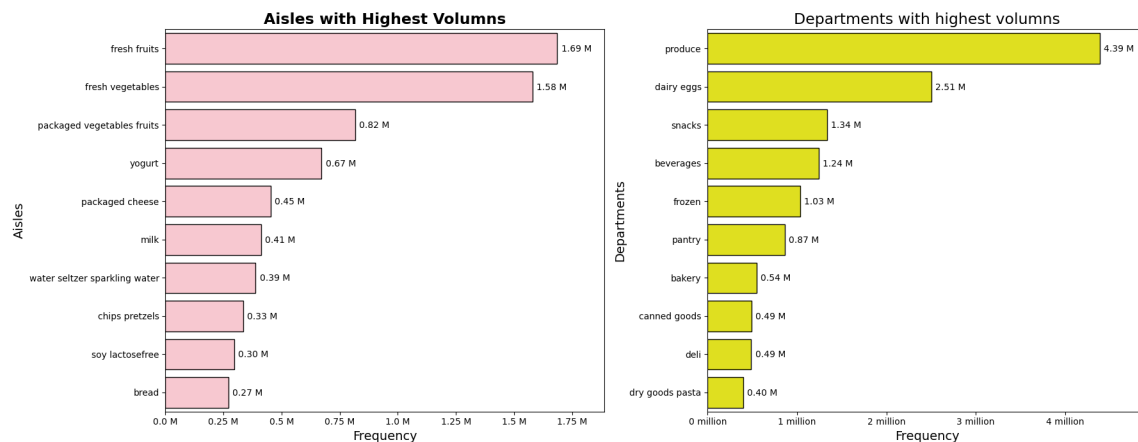
### Most Ordered Products



#### INSIGHT & RECOMMENDATION

Bananas are, by a wide margin, the single most ordered item on the platform at 219,146 orders, with 9 of the top 10 products being produce. **this means produce needs to be treated as the anchor category**, not just another aisle, a stockout here costs the business more than almost anywhere else.

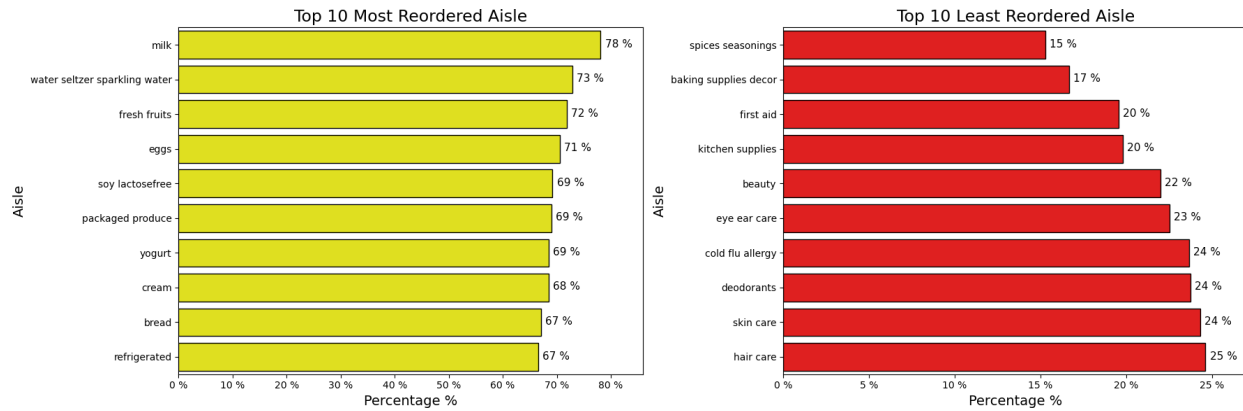
### Departments and Aisles Driving Volume



#### INSIGHT & RECOMMENDATION

Produce brings in 4.39 million items, almost double Dairy Eggs at 2.51 million, and it's mostly Fresh Fruits and Fresh Vegetables driving that number. **if extra warehouse space or better supplier terms are being negotiated**, produce is where that conversation should start.

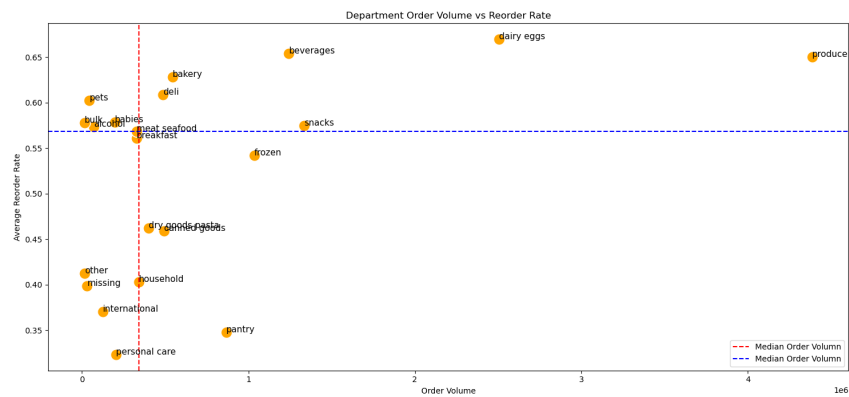
## Reorder Rate by Aisle



### INSIGHT & RECOMMENDATION

Milk, water, fresh fruits, and eggs top the reorder charts, all above 70%, while spices and baking supplies barely reorder at all, around 15-17%. **reorder rate tracks how fast a product gets used up**, not its price or popularity, so subscription and auto-reorder features should be pushed hard on the first group and not the second.

## Volume vs. Loyalty by Department

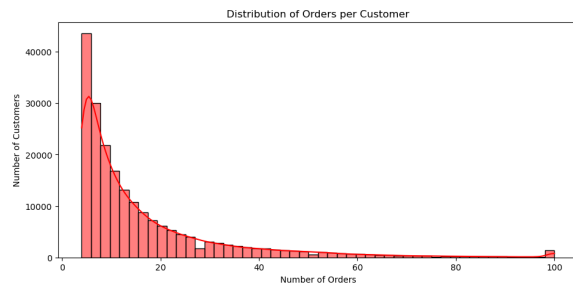


### INSIGHT & RECOMMENDATION

High-volume departments are not just riding one-time curiosity purchases. Produce, Dairy Eggs, and Beverages are simultaneously the highest-volume and the highest-loyalty categories. **the recommendation here is simple:** keep investing in these three, the loyalty backs up the sales.

## How Often and How Much

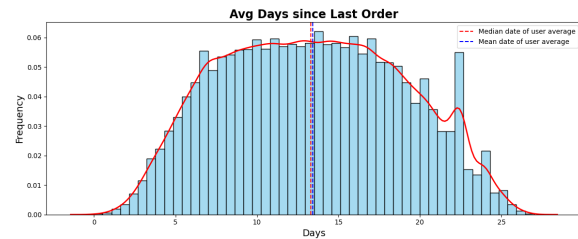
## Orders per Customer



### INSIGHT & RECOMMENDATION

Most customers are light users — the largest group has placed only 4 to 6 total orders. The small tail near 100 is most likely a dataset recording cap, not real behavior. **retention spend has more to gain** from occasional shoppers than the already-loyal few.

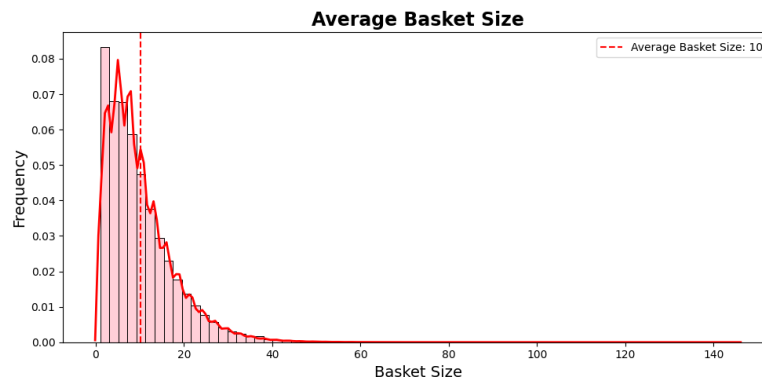
## Days Between Orders



### INSIGHT & RECOMMENDATION

Customers reorder every 13 days on average, but that hides two rhythms: a weekly group and a near-monthly group around 22-23 days. **a single reminder schedule won't fit both**; two separate cadences would serve this better.

## Average Basket Size

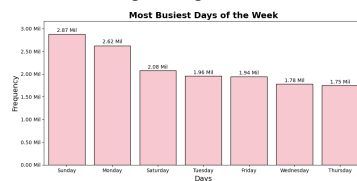


### INSIGHT & RECOMMENDATION

The average basket size is 10 items, but that figure is skewed upward by a small number of large orders. Most orders actually contain just 1 to 5 items. **fulfillment and packaging should be planned around the smaller, far more common basket size, not the average.**

## When Demand Peaks

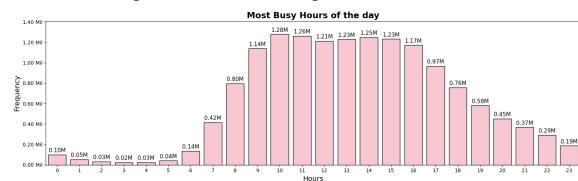
### Orders by Day of Week



### INSIGHT & RECOMMENDATION

Sunday and Monday are clearly the busiest days; Saturday actually ranks third. **delivery staff and restocking should be scheduled around Sunday and Monday specifically**, not around “the weekend” broadly.

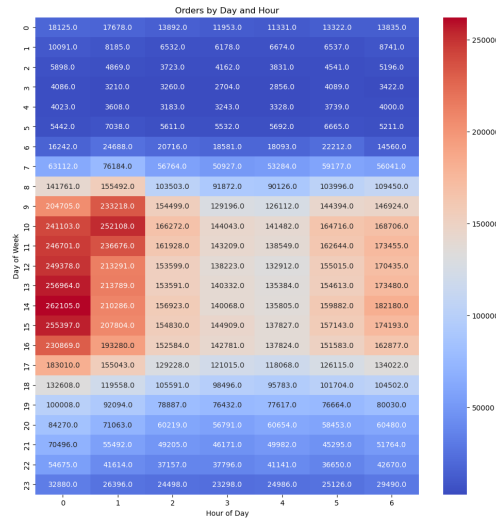
### Orders by Hour of Day



### INSIGHT & RECOMMENDATION

Order volume plateaus at its peak from 10 AM to 4 PM rather than spiking at one hour. **staffing should reflect that full window**, not a short lunchtime rush.

## Day and Hour Combined

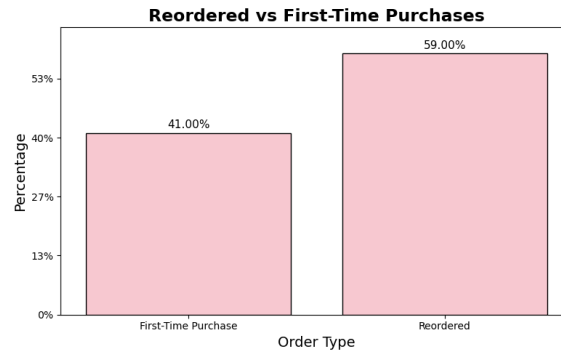


### INSIGHT & RECOMMENDATION

The two effects compound: the busiest slot overall is Sunday/Monday combined with 10 AM to 4 PM, clearing 260,000 orders — higher than either factor predicts alone. **staffing plans should use this combined view**, not two separate day-only and hour-only schedules.

## Repeat Behavior and Loyalty

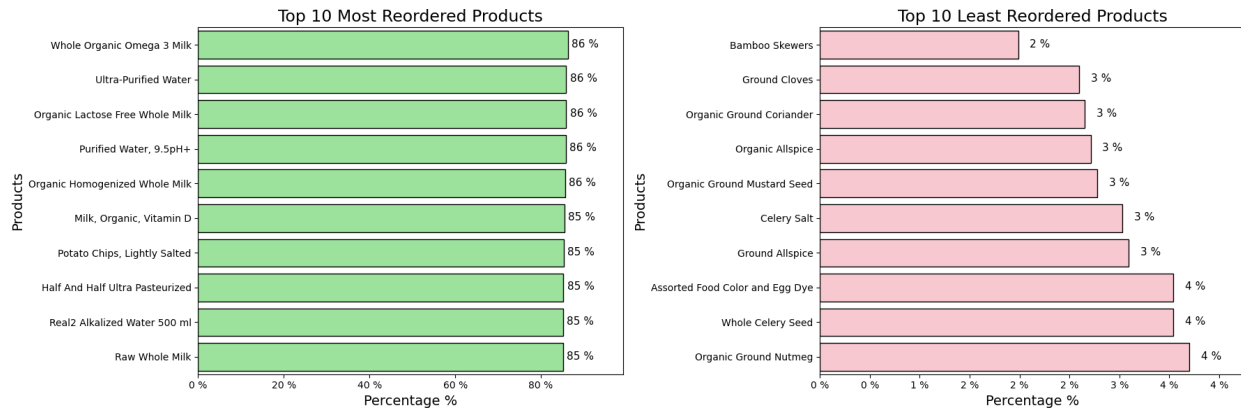
### Reorders vs. First-Time Purchases



### INSIGHT & RECOMMENDATION

59% of everything ordered is a reorder, only 41% is a first-time purchase. **retention already has a strong base to build on**, but new customer acquisition needs its own separate push — it won't happen on the back of reorders alone.

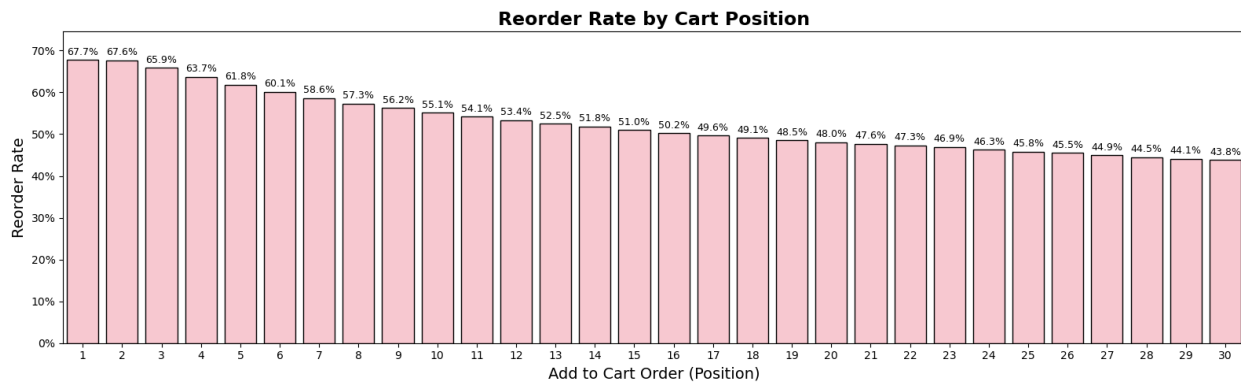
## Products With the Highest and Lowest Reorder Rates



### INSIGHT & RECOMMENDATION

Milk and water variants sit at 85-86% reorder rate; spice and specialty baking items fall to 2-4%. **loyalty and subscribe programs should concentrate on the milk/water tier**; that budget wouldn't move the needle on low-frequency items like ground cloves.

## Reorder Likelihood by Cart Position

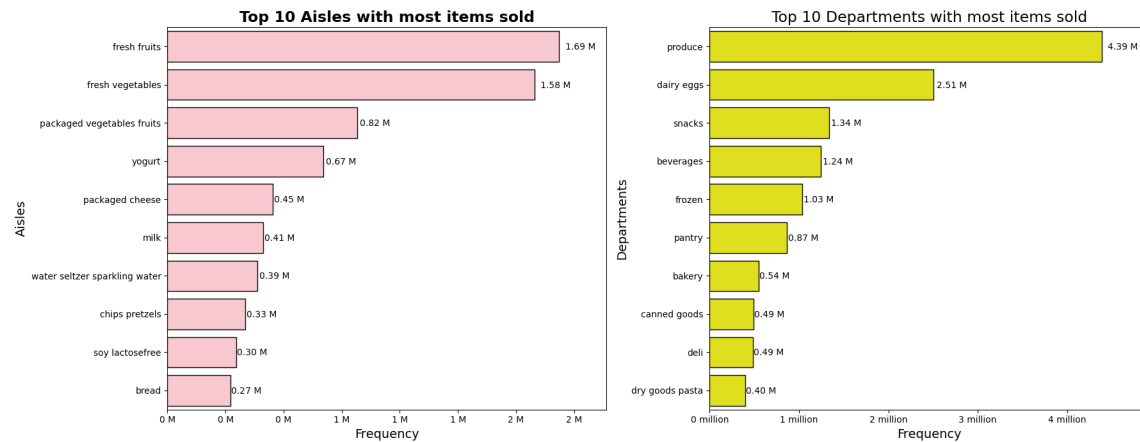


### INSIGHT & RECOMMENDATION

Reorder likelihood drops steadily from 67.7% on the first item added to 43.8% by the thirtieth, showing customers add routine items first and new items later in a session. **in-app suggestions should mirror this**: routine reorders surfaced early, discovery items placed later in the flow.

## Category Priorities

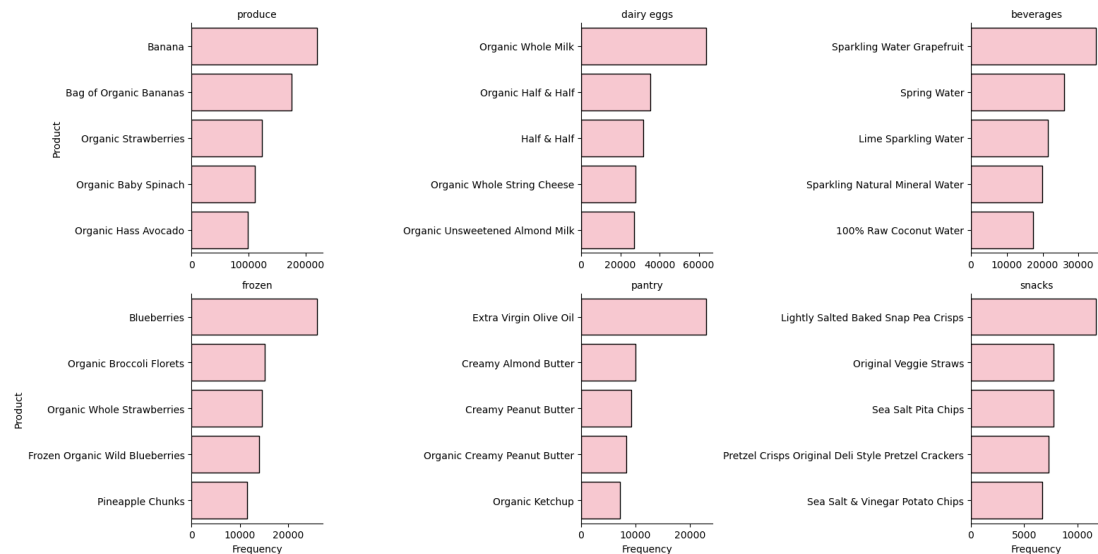
## Top Categories by Total Items Sold



### INSIGHT & RECOMMENDATION

This confirms the earlier pattern: Produce leads everything at 4.39 million items, with Fresh Fruits and Fresh Vegetables carrying most of that total. **there isn't a case for treating any other department as equally important** right now.

## Top Products Within Each Leading Department



### INSIGHT & RECOMMENDATION

Most departments are carried by one dominant product — Banana in Produce, Organic Whole Milk in Dairy Eggs, Olive Oil in Pantry. Snacks is the exception, with a much more even spread across its top 5. **supply reliability should concentrate on those single dominant products**, while Snacks is better suited to a rotating, discovery-driven approach.

## Recommendations

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Pulling the findings above together, these are the actions that follow most directly from the data, in the order they would have the most impact:

- Protect produce and dairy availability first. They carry both the highest volume and the highest reorder rates, so a stockout in either category costs more than anywhere else on the platform.
- Build subscription or auto-reorder features specifically around milk, water, and produce, and skip that investment on low-reorder categories like spices, where the underlying behavior won't support it.
- Plan staffing and delivery capacity around the Sunday-Monday, 9 AM to 4 PM window specifically, rather than spreading resources evenly across the week.
- Run two separate reorder-reminder cadences, roughly weekly and roughly every three weeks, to match the two customer rhythms found in the order-frequency data.
- Redesign in-app product suggestions to surface routine items early in the shopping session and discovery items later, matching how customers actually build their carts.
- Focus retention spend on the broad base of light and occasional customers rather than the already-loyal segment, since that is where most of the customer base, and most of the untapped growth, currently sits.